

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250268400

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Rodriguez; Adrian et al.

REDUCING POWER CONSUMPTION OF REFRIGERATED DISPLAY CASE WITH ELECTRONIC DOOR DISPLAY

Abstract

Methods and systems are disclosed for reducing power consumption of refrigerated display cases with electronic door displays. A door system for a refrigerated display case includes: a movable door configured to provide access to the refrigerated display case; an electronic display mounted to the movable door; a sensor; at least one processor in electronic communication with the sensor and the electronic display; and a data store coupled to the at least one processor having instructions stored thereon which, when executed by the at least one processor, causes the at least one processor to perform operations comprising: receiving, from the sensor, data indicating a position of the movable door; and in response to receiving the data indicating the position of the movable door, adjusting a power setting of the electronic display.

Inventors: Rodriguez; Adrian (Santa Clarita, CA), Almaguer; Pedro (North Hills, CA), Rivera; Francisco Flores (Valencia, CA)

Applicant: Anthony, Inc. (Sylmar, CA)

Family ID: 1000007715439

Appl. No.: 18/588331

Filed: February 27, 2024

Publication Classification

Int. Cl.: A47F3/04 (20060101)

U.S. Cl.:

CPC A47F3/0478 (20130101); A47F3/043 (20130101);

Background/Summary

TECHNICAL FIELD

[0001] This invention relates to thermally insulated doors for temperature-controlled display cases.

BACKGROUND

[0002] Refrigerated display cases are used in commercial, institutional, and residential applications for storing and/or displaying refrigerated or frozen objects. Refrigerated display cases may be maintained at temperatures above freezing (e.g., a refrigerator) or at temperatures below freezing (e.g., a freezer). Refrigerated display cases have one or more thermally insulated doors or windows for viewing and accessing refrigerated or frozen objects within a temperature-controlled space. Doors for refrigerated display cases generally include thermally insulated glass panel assemblies. Displaying content on refrigerated display case doors can be an effective way of advertising products. Refrigerated display case doors with embedded display panels consume electrical power.

SUMMARY

[0003] The present disclosure relates to reducing power consumption of a refrigerated display case with an electronic door display. The display case door includes electrical components such as an electronic display, a heating assembly, a camera, and lighting. The amount of power consumed by the electrical components can be reduced by reducing the amount of time that the electrical components are on and/or by switching the electrical components to a reduced power mode.

[0004] Implementations of the present disclosure include a refrigerated display case, including: a movable door configured to provide access to the refrigerated display case; an electronic display mounted to the movable door; a sensor configured to indicate one or more positions of the movable door; at least one processor in electronic communication with the sensor and the electronic display; and a data store coupled to the at least one processor having instructions stored thereon which, when executed by the at least one processor, causes the at least one processor to perform operations including: receiving, from the sensor, data indicating a position of the movable door; and in response to receiving the data indicating the position of the movable door, adjusting a power setting of the electronic display.

[0005] In some implementations, adjusting the power setting of the electronic display includes turning the electronic display off or on.

[0006] In some implementations, the refrigerated display case includes a circuitry module including a media player in electronic communication with the electronic display, the media player configured to control media content presented on the electronic display.

[0007] In some implementations, adjusting the power setting of the electronic display includes transmitting an instruction to the circuitry module to present no content on the electronic display.

[0008] In some implementations, the refrigerated display case includes a light configured to illuminate an interior of the refrigerated display case, the operations including: in response to receiving the data indicating the position of the movable door, adjusting a power setting of the light.

[0009] In some implementations, adjusting the power setting of the light includes turning the light off or on.

[0010] In some implementations, the data indicating the position of the movable door includes data indicating that the movable door is shut; and adjusting the power setting of the light includes turning the light off after a delay period.

[0011] In some implementations, the refrigerated display case includes a heater configured to provide heat to at least a part of the refrigerated display case, the operations including: in response to receiving the data indicating the position of the movable door, adjusting a power setting of the heater.

[0012] In some implementations, adjusting the power setting of the heater includes turning the heater off or on.

[0013] In some implementations, the heater includes an anti-condensation heater configured to provide heat to the movable door.

[0014] In some implementations, the data indicating the position of the movable door includes data indicating that the movable door is open or shut.

[0015] In some implementations, the data indicating the position of the movable door includes data indicating that the movable door has been open for at least a threshold time duration.

[0016] In some implementations, the data indicating the position of the movable door includes data indicating a swing angle of the movable door.

[0017] In some implementations, an internal volume of the refrigerated display case holds an inventory, and the electronic display is configured to present a planogram representing the inventory.

[0018] In some implementations, the movable door includes an insulated panel.

[0019] In some implementations, the sensor includes at least one of a camera, a proximity sensor, a contact sensor, a position switch, a rheostat, an accelerometer, a gyroscope, an infrared sensor, a radar sensor, and an ultrasonic sensor.

[0020] Further implementations of the present disclosure include a refrigerated display case system, including: a case including an internal volume; and a door system including: a movable door configured to provide access to the internal volume of the case; an electronic display mounted to the movable door; a sensor configured to indicate one or more positions of the movable door; at least one processor in electronic communication with the sensor and the electronic display; and a data store coupled to the at least one processor having instructions stored thereon which, when executed by the at least one processor, causes the at least one processor to perform operations including: receiving, from the sensor, data indicating a position of the movable door; and in response to receiving the data indicating the position of the movable door, adjusting a power setting of the electronic display.

[0021] In some implementations, adjusting the power setting of the electronic display includes turning the electronic display off or on.

[0022] In some implementations, the refrigerated display case includes a circuitry module including a media player in electronic communication with the electronic display, the media player configured to control media content presented on the electronic display.

[0023] Further implementations of the present disclosure include a computer-implemented method of operating a refrigerated display case, including: receiving, by one or more computers and from a sensor, data indicating a position of a movable door configured to provide access to the refrigerated display case; and in response to receiving the data indicating the position of the movable door, adjusting, by the one or more computers, a power setting of an electronic display mounted to the movable door.

[0024] Particular implementations of the subject matter described in this specification can be implemented so as to realize one or more of the following advantages.

[0025] Implementations of the present disclosure may improve the power efficiency of refrigerated display cases with electronic displays mounted to the display case door. For example, implementations of the present disclosure may provide for reducing power consumption of an electronic display by turning the electronic display off when the door is open to a particular swing angle and/or when the door is open for at least a threshold time duration, such as when the display case is being restocked. Power consumption can also be reduced by turning lights off inside the display case when the door is shut. The lights may shut off after a short delay to ensure sufficient light for any inventory tracking cameras to capture an accurate picture of the products prior to cycling the lights off. Power consumption can also be reduced by turning a door heater off when the door is open.

[0026] The details of one or more embodiments of the invention are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the invention will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

[0027] FIGS. **1-2** show an exemplary refrigerated display case according to some implementations of the present disclosure.

[0028] FIG. **3** shows an exploded perspective view of an exemplary arrangement of an electronic display assembly in a display case door according to implementations of the present disclosure.

[0029] FIG. **4** shows a perspective view of an exemplary electronic display case door with the electronic display assembly mounted in door frame with bezels installed according to implementations of the present disclosure.

[0030] FIG. **5** shows an exemplary display case door and control circuitry according to implementations of the present disclosure.

[0031] FIG. **6A** shows a top view of an exemplary display case with the door in the closed position according to implementations of the present disclosure.

[0032] FIG. **6B** shows a top view of an exemplary display case with the door in the open position according to implementations of the present disclosure.

[0033] FIG. **7** shows an example control system for a refrigerated display case according to implementations of the present disclosure.

[0034] FIG. **8** shows a flow chart of an example process for reducing power consumption of a refrigerated display case with an electronic door display by adjusting a display power setting according to implementations of the present disclosure.

[0035] FIG. **9** shows a flow chart of an example process for reducing power consumption of a refrigerated display case with an electronic door display by adjusting a lighting setting according to implementations of the present disclosure.

[0036] FIG. **10** shows a flow chart of an example process for reducing power consumption of a refrigerated display case with an electronic door display by adjusting a heater setting according to implementations of the present disclosure.

DETAILED DESCRIPTION

[0037] FIGS. **1-2** show an exemplary refrigerated display case system **10**. The refrigerated display case system **10** can be a refrigerator, freezer, or other enclosure defining a temperature-controlled space. In some implementations, refrigerated display case system **10** is a refrigerated display case. For example, refrigerated display case system **10** can be a refrigerated merchandiser in retail establishments such as grocery stores, supermarkets, convenience stores, florist shops, and/or other commercial settings to store and display temperature-sensitive consumer goods (e.g., food products and the like). The refrigerated display case system **10** can be used to display products that must be stored at relatively low temperatures and can include shelves, glass doors, glass walls, and/or video displays to permit viewing of the products supported by the shelves. In some implementations, refrigerated display case system **10** is a refrigerated storage unit used, for example, in warehouses, restaurants, and lounges. Refrigerated display case system **10** can be a free standing unit or “built in” unit that forms a part of the building in which refrigerated display case system **10** is located.

[0038] The refrigerated display case system **10** includes a container **12**. Container **12** includes a top wall **14**, a bottom wall **16**, a left side wall **18**, a right side wall **20**, a rear wall (not shown), and a front portion **22** defining a temperature-controlled space. The container **12** encloses an internal volume of the refrigerated display case. The internal volume of the container **12** is configured to hold an inventory of products.

[0039] Front portion **22** includes an opening into the temperature-controlled space. A thermal frame **24** can be mounted at least partially within the opening. The thermal frame **24** includes a plurality of perimeter frame segments (i.e., a header or top frame segment **26**, a sill or bottom frame segment **28**, a left side frame segment **30**, and a right side frame segment **32**) forming a closed shape along a perimeter of the opening. In some implementations, the thermal frame **24** includes one or more mullion frame segments **34** dividing the opening into multiple smaller openings. For example, FIG. **1** illustrates a three-door assembly with a pair of mullion frame segments **34** extending between top frame segment **26** and bottom frame segment **28** to divide the opening into three smaller openings. Each of the smaller openings may correspond to a separate display case door **100** of the three-door assembly. In other implementations, mullion frame segments **34** may be omitted. For example, FIG. **2** illustrates a one-door assembly in which thermal frame **24** includes perimeter frame segments **26-32** but not mullion frame segments **34**. In some implementations, thermal frame **24** includes top frame segment **26** and bottom frame segment **28** with no side frame segments **30** or **32**. In such implementations, thermal frame **24** may include one or more mullion frame segments (e.g., such as mullion frame segment **34** shown in FIG. **1**) depending, for example, on the size of the refrigerated enclosure in which thermal frame **24** is to be installed and the number of doors.

[0040] Refrigerated display case system **10** includes one or more movable doors. The movable doors are configured to provide access to the internal volume of the refrigerated display case system **10**. In some examples, display case doors **100** are pivotally mounted on the thermal frame **24** by hinges **38**. In some implementations, the display case doors **100** are sliding doors configured to open and close by sliding relative to the thermal frame **24**. The example display case doors **100** illustrated in FIGS. **1** and **2** include panel assemblies **40** and handles **42**. Referring to FIG. **2**, thermal frame **24** includes a series of contact plates **44**. Contact plates **44** are attached to a front surface of thermal frame **24** and provide a sealing surface against which display case doors **100** rest in the closed position. For example, doors **100** may include a gasket or other sealing feature around a perimeter of each display case door **100**. The gaskets may employ a flexible bellows and magnet arrangement, which, when the display case doors **100** are closed, engage contact plates **44** to provide a seal between doors **100** and thermal frame **24**. The thermal frames provide a thermally conductive path from the frame segments **26-32**, for maintaining the temperature of the contact plates **44** at or close to the temperature of the external environment (e.g., the environment outside of the refrigerated display case system **10**) and to aid in preventing condensation from forming on the contact plates **44**. Preventing condensation on the contact plates may provide for a more positive seal between the contact plates **44** and a magnetic gasket on the door, thereby improving the thermal properties of the refrigerated display case system **10**.

[0041] FIG. **3** is an exploded perspective view of an exemplary arrangement of an electronic display assembly **102** in a display case door **100** according to implementations of the present disclosure. FIG. **4** illustrates an exemplary display case door **100** that can be installed in a refrigerated display case such as a refrigerator, a freezer, or other enclosure defining a temperature-controlled space. Display case door **100** includes an insulated panel assembly **104** or transparent panel, a door frame **106** secured to an edge of insulated panel assembly **104**, and mounting blocks **108**. Mounting blocks are coupled to door frame **106**. The display case door **100** include an electronic display assembly **102** that is coupled to door frame **106** by way of mounting blocks **108**. Electronic display assembly **102** can be mounted to door frame **106** so as to overlay all or a majority of insulated panel assembly **104**. Electronic display assembly **102** includes electronic display **110**. The electronic display assembly **102** can be mounted in front of an insulated panel assembly (e.g., a glass insulated panel assembly or an opaque foam panel assembly) in a door frame secured to an edge of the insulated panel assembly. The electronic display assembly **102** can include, e.g., a video display in electronic communication with control circuitry configured to display video and/or images including, but not limited to, images of products contained in a display

case, advertisements, and nutritional information. In some examples, the electronic display is configured to present a planogram representing inventory held within the internal volume of the refrigerated display case.

[0042] In some implementations, a display case door includes one or more bezels. The bezels can cover all or a portion of the mounting blocks and other components for mounting the electronic display assembly **102** to the movable door. The bezels can create a recognized UL wireway.

[0043] Insulated panel assembly **104** can include one or more panes of glass. In some implementations, insulated panel assembly **104** includes two or more layers of transparent panes bounding a sealed space in between, forming a sealed glass unit (SGU). Door frame **106** extends around and is secured to a peripheral edge of insulated panel assembly **104**. As further described in detail below, door frame **106** defines a channel or tunnel that receives one or more power cables that provide electrical power to the electronic display assembly.

[0044] Display case door **100** can include a single electronic display or multiple electronic displays. For example, display case door **100** can include two or more electronic displays vertically stacked and together covering the insulated panel assembly **104**.

[0045] Electronic display **110** can include, but is not limited to, a liquid crystal display (LCD), a light emitting diode (LED) display, an organic light emitting diode (OLED) display, a field emission display (FED), a plasma display panel (PDP), or an electroluminescent (EL) display. For example, electronic display **110** can be a smart television with streaming capabilities for receiving content over a wireless network (e.g., a Wi-Fi network). Electronic display **110** is generally opaque and, when mounted to the movable display case door **100** partially or completely obstructs the view through the insulated panel assembly **104**. In some implementations, electronic display **110** can be a custom sized display configured to correspond with dimensions of the display case door **100**. In some implementations, electronic display **110** can be a commercial off the shelf (COTS) display.

[0046] FIG. 4 is a perspective view of the electronic display case door **100** of FIG. 3 with electronic display assembly **102** mounted in door frame **106**, with bezels installed around the edges of door frame **106**. Display case door **100** includes left bezel **124**, top bezel **126**, and right bezel **128**. Each of left bezel **124**, top bezel **126**, and right bezel **128** overlay and conceal a portion of mounting blocks **108**, left side hinge brackets **112**, right side brackets **114**, hangers **116**, and display supports **118**.

[0047] Display case door **100** includes handle **130**. Electronic display assembly **102** includes circuitry module **132**, top sensor **134**, and handle-side sensor **136**. The outside edges of circuitry module **132** are secured to left rail **206** and right rail **208** of door frame **106**. Left bezel **124** and right bezel **128** can be attached to their respective rails. Top sensor **134** passes through an opening or cutout in top bezel **126**. Handle-side sensor **136** is accessible through a corresponding aperture in right bezel **128**.

[0048] Hold open bracket **138** is provided on a top edge of display case door **100**. One end of hold open bracket **138** is pivotally coupled to door frame **106**. In service, hold open bracket **138** can be used to maintain door in a desired open position.

[0049] Circuitry module **132** can be positioned in the door **100** itself. For example, circuitry module **132** can be positioned in a bottom portion of door frame **106**. Circuitry module **132** overlays a portion of insulated panel assembly **104**. Circuitry module **132** can be attached to electronic display **110** such that, with electronic display **110** releasably coupled to door frame **106**, circuitry module **132** is releasably coupled to door frame **106**. Circuitry module **132** can include a media player in electronic communication with electronic display **110** to control media content presented on electronic display **110**.

[0050] In various implementations, door frame **106** has a width and thickness that allows display case door **100** to be installed and operated in an existing refrigerated display case without the need of retrofitting the display case. Display case door **100** can be operated with or without electronic display **110**.

[0051] FIG. 5 illustrates an exemplary display case door **100** according to an implementation of the present disclosure. In this example, display case door includes quick disconnect plug **535**. Quick disconnect plug **535** is electrically coupled to a power cable **525**, a ground wire **527**, and one or more electrical wires **529** coupled to electronic display **510** or to circuitry module **526**. In some implementations, power cable **525** includes at least a power supply cable, a ground cable, and a detachable ground lug in one wiring assembly with one plug on each end of the wiring assembly. For example, the electronic display **510** can have a male component while the door frame **502** can have the female component. The male component and the female component can be referred to as a pin and socket connection. The plug will have all the power, ground and neutral joined at the plug.

[0052] The display case door **100** includes a transparent panel assembly **522** and a door frame **502** that receives and secures an electronic display **510** to cover transparent panel assembly **522**. Electronic display **510** has a height that is less than the interior height of frame **502**, leaving a space **528** between the bottom of door **100** and the bottom edge of the electronic display **510** to place electrical components with door **100**. For example, display case door **100** includes circuitry module **526** disposed under the electronic display **510**. Circuitry module **526** includes at least a media player **530** in electronic communication with electronic display **510**. Media player **530** controls media content presented on display **510**. Circuitry module **526** can also include a power hub **532** and a power converter **534**. Circuitry module **526** is releasably coupled to door frame **502**.

[0053] Power cables **525** can provide electrical power to electronic display **510** and power converter **534**. Power converter **534** receives alternating current (AC) power from power cable **525** and converts the AC power to direct current (DC) power. For example, power converter **534** converts 110V AC power to 5V DC power. Power converter **534** is electrically connected to one or more media players **530**. Power converter **534** is arranged inside display case door **100**.

[0054] FIG. 6A illustrates a top view of a refrigerated display case system **10** including a display case **11** with a movable door **100** in the closed position. FIG. 6B illustrates a top view of the refrigerated display case system including the display case **11** with the movable door **100** in the open position. As shown in FIGS. 6A and 6B, display case **11** can be opened by applying force to door handle **15**, causing door **100** to pivot about hinge **29**. In other implementations, display case **11** can be opened by sliding door **100** with respect to the case frame.

[0055] The refrigerated display case system **10** can include a lighting system including lights **64a**, **64b** ("lights **64**"). The lights **64** can be configured to illuminate the internal volume of the display case **11**. In some examples, the lights **64** are located inside the internal volume of the display case **11**. In some examples, the lights **64** are located outside of the internal volume of the display case **11**. In some examples, the lights **64** are attached to the door **100**. In some examples, the lights **64** are communicably coupled to one or more sensors **68** and/or one or more cameras **65**. For example, the lights **64** can receive a signal from the sensors **68**, the cameras **65**, or both, indicating when the door **100** is open and/or shut. In some examples, the lights **64** are operated by a control system such as the control system **400** shown in FIG. 7. The control system **400** can operate the lights **64** in order to turn the lights **64** on, to turn the lights **64** off, and to adjust brightness of the lights **64**. In some examples, the control system **400** can operate each light independently. For example, the control system **400** can operate the light **64a** independently from the light **64b**.

[0056] The refrigerated display case system **10** can include an alarm **62**. The alarm **62** can be, for example, an audible alarm, a visual alarm, or both. The alarm **62** can be configured to activate in order to notify personnel of an anomaly. For example, the alarm **62** can be configured to activate in order to notify personnel that the door **100** has been open for greater than a threshold time duration. In some examples, the alarm **62** is communicably coupled to the sensors **68** and/or the camera **65**. For example, the alarm **62** can receive a signal from the sensors **68** indicating when the door **100** is open and/or shut. In some examples, the alarm **62** is operated by a control system such as the control system **400** shown in FIG. 7. The control system **400** can operate the alarm **62** in order to turn the alarm **62** on and to turn the alarm **62** off. In some examples, the alarm **62** is located at or

near the display case **11**. In some examples, the alarm **62** is located away from the display case **11**. For example, the alarm **62** can be located in a manager's office or in a storage room of a retail establishment. In some examples, the refrigerated display case system includes multiple alarms. For example, the refrigerated display case system can include an alarm located at or near the display case **11**, and another alarm located in a manager's office.

[0057] The refrigerated display case system **10** can include a heating system including a heater **66**. The heater **66** can be configured to provide heat to at least part of the refrigerated display case **11**. In some examples, the heater **66** is an anti-condensation heater that is configured to provide heat to the movable door **100**. In some examples, the heater **66** is communicably coupled to the sensors **68** and/or the camera **65**. For example, the heater **66** can receive a signal from the sensor **68**, the camera **65**, or both indicating when the door **100** is open and/or shut. In some examples, the heater **66** is operated by a control system such as the control system **400** shown in FIG. 7. The control system **400** can operate the heater **66** in order to turn the heater **66** on, to turn the heater **66** off, and to adjust a power setting of the heater **66**. For example, the heater **66** may have adjustable power settings ranging between lower heat output and higher heat output.

[0058] The refrigerated display case system **10** can include a sensor **68** that is configured to detect movement of door **100** from a closed position (as depicted in FIG. 6A) to an open position (as depicted in FIG. 6B). In some implementation, the sensors **68** generates a signal indicating a swing angle **60** of the door **100**.

[0059] The sensors **68** can include, but are not limited to, a contact sensor (e.g., contact switch), a movement sensor (e.g., accelerometer, gyroscope, rheostat), a proximity sensor (e.g., optical sensor, infrared sensor, ultrasonic sensor, capacitive sensor, inductive sensor, magnetic sensor), or any combination thereof. In some examples, the sensors **68** includes multiple different types of sensors integrated into a single device.

[0060] Although shown as being located at the top corner of the door **100**, the sensors **68** can be located at other locations. For example, the sensors **68** can be attached at another position of the door such a bottom corner or at an edge of the door **100** away from a corner, or coupled to the hinge **29**. In some examples, the sensors **68** is attached to a surface of the door **100**. In some examples, the sensors **68** is attached to the display case **11**. In some examples, the refrigerated display case system includes multiple sensors. The multiple sensors can be co-located or can be located in multiple different locations of the refrigerated display case system.

[0061] In some implementations, the sensors **68** can detect conditions including, but not limited to, light levels, temperature, and humidity levels. The sensors **68** can be configured to share data with media players and/or personal computers through a universal serial bus (USB) connection. In some implementations, the sensors **68** is attached to a device that has a wireless Internet connection (i.e., "Wi Fi enabled" device), and can send signals to remote computing devices using the wireless Internet connection.

[0062] In some implementations, sensors **68** can be communicably coupled to a control system, such as control system **400** shown in FIG. 7. The control system **400** can include a processor or processors in communication with the sensors **68** and with the display **110**. The control system **400** can include a data store coupled to the processor. The data store can store instructions. When executed by the processor, the instructions cause the processor to perform operations.

[0063] In some implementations, the control system **400** can include instructions to perform automatic actions in response to receiving data from sensor **68**. For example, the control system **400** can receive data from the sensor indicating a position of the movable door. The data indicating the position of the movable door can include data indicating that the movable door is open. The data indicating the position of the movable door can include data indicating that the movable door is shut. In some examples, the data indicating the position of the movable door can include data indicating a direction of motion of the movable door. For example, the data can indicate that the movable door is swinging open or is swinging shut. In some examples, the data indicating the

position of the movable door can include data indicating a swing angle of the movable door (e.g., an amount by which the movable door is open). For example, the data can indicate that the movable door partially open, is fully open, or is propped open to a hold-open position. In some examples, the data indicating the position of the movable door includes data indicating a time that the movable door has been open. In some examples, the data indicating the position of the movable door includes data indicating that the movable door has been open for at least a threshold open time duration.

[0064] In response to receiving the data from the sensor, the control system **400** can adjust a power setting of one or more components of the display case **11**. In some examples, the control system **400** can adjust a power setting of the display **110**. In some examples, in addition to or instead of adjusting the power setting of the display **110**, the control system **400** can adjust a power setting of lighting, heating, or both. In some examples, in addition to or instead of adjusting the power setting of the display **110**, the control system **400** can activate or deactivate an alarm.

[0065] In some examples, the control system **400** is configured to gather and analyze door operating metrics related to opening of display case door **100**. The control system **400** can include instructions to perform data analytics and record door operating metrics about customer interaction with the display case each time the system receives a signal from sensor **68**. These metrics can include, but are not limited to, the time at which display case door **100** is opened, the total number of times display case door **100** is opened in a given time period, or the frequency of openings of door **100** for a given time period.

[0066] In some implementations, at least a majority of shelves contained within display case **11** are within the field of view of camera **65** when door **100** is open by at least a specified swing angle **60**. Sensors **68** can be communicably coupled to camera **65**. For example, sensors **68** can transmit a signal to camera **65** to initiate camera **65** to capture images of the interior of display case **11** when sensors **68** detects that display case door **100** is in an open position.

[0067] In some examples, the sensors **68** includes a contact sensor **68a**. The contact sensor **68a** can include a contact switch that detects whether or not there is contact between the door **100** and a component of the display case **11**. For example, the contact sensor can detect contact between an edge of the door **100** and a door frame of the display case **11**.

[0068] In some examples, the sensors **68** includes a proximity sensor **68b**. The proximity sensor **68b** can detect and/or measure a proximity between the location of the proximity sensor **68b** and a component of the display case **11** such as a door frame. For example, with the proximity sensor **68b** attached to the door **100**, the measured proximity between the proximity sensor **68b** and the door frame can indicate an amount by which the door **100** is open. The proximity sensor **68b** can include, for example, an optical sensor, an infrared sensor, an ultrasonic sensor, a capacitive sensor, an inductive sensor, a magnetic sensor, or any combination of these.

[0069] In some examples, the sensors **68** includes a movement sensor **68c**. The movement sensor **68c** can detect and/or measure movement of the movement sensor **68c**. For example, with the movement sensor **68c** attached to the door **100**, the measured movement of the movement sensor **68c** can indicate a direction and speed of movement of the door **100** as the door opens and shuts. The movement sensor **68c** can include, for example, an accelerometer or a gyroscope.

[0070] The control system **400** can receive, as input, sensor data from the sensors **68**. For example, the control system **400** can receive contact information from the contact sensor **68a**, proximity information from the proximity sensor **68b**, movement information from the movement sensor **68c**, or any combination thereof. In some examples, the control system **400** receives, as input, images captured by the camera **65**.

[0071] The control system **400** can make one or more determinations based on the input. In some examples, the control system **400** processes the input data to determine a status of the door **100**. For example, the control system **400** can determine, based on the input data, that the door **100** is open or that the door **100** is shut. In some examples, the control system **400** can determine an amount of

time that the door **100** has been open or an amount of time that the door **100** has been shut. In some examples, the control system **400** can determine a swing angle of the door **100** and/or a direction of motion of the door **100**.

[0072] The control system **400** can provide, as output, control signals that instruct components of the refrigerated display case system **10** to perform one or more actions. In some examples, the control system **400** outputs control signals to the alarm **62**. The control signals can turn the alarm **62** on or off. The alarm **62** can provide an indication to personnel of the status of the door **100**. For example, the alarm **62** can indicate that the door **100** has been opened for a threshold open time duration. Activating the alarm **62** can cause a reduction in power consumption by alerting personnel to anomalies in door operation, enabling the personnel to investigate and remediate the anomalies. For example, a customer may accidentally open the door **100** to the hold-open position. The control system **400** can activate the alarm **62** when the door **100** has been in the hold-open position for the threshold open time duration, alerting personnel who can then shut the door **100**, conserving energy.

[0073] In some examples, the control system **400** outputs control signals to the display **110**. The control signals can adjust a power setting of the display **110**. For example, the control signals can turn the display **110** off, turn the display **110** on, or reduce or increase a brightness of the display **110**. For example, the control system **400** can control operations of the circuitry module **526** installed in the door **100** and/or the control system **400** can control operations of the media player **530**. For example, the control system **400** can send analog or digital signals to the circuitry module **526** or the media player **560** to control operations or override operations of either. In some implementations, the control system **400** can be included as a sub-module of the circuitry module **526**.

[0074] In some examples, the control system **400** outputs control signals to the lights **64**. The control signals can adjust a power setting of the lights **64**. For example, the control signals can turn the lights **64** on, turn the lights **64** off, or reduce or increase a brightness of the lights **64**.

[0075] In some examples, the control system **400** outputs control signals to the heater **66**. The control signals can adjust a power setting of the heater **66**. For example, the control signals can turn the heater **66** on, turn the heater **66** off, or reduce or increase heat output by heating elements of the heater **66**.

[0076] FIG. **8** shows a flow chart of an example process **800** for reducing power consumption of a refrigerated display case with an electronic door display by adjusting a display power setting. The process **800** can be performed by a computing system including one or more computers, e.g., control system **400** or circuitry module **526**. The order of steps in the process **800** is illustrative only, and the steps can be performed in different orders and/or in parallel. In some implementations, the process **800** can include additional steps, fewer steps, or some of the steps can be divided into multiple steps. In some examples, the steps of the process **800** can be performed by different components of the disclosed systems.

[0077] The process **800** includes receiving data indicating that a door of a refrigerated display case is open (**802**). For example, the control system **400** can receive data from the sensors **68**, the camera **65**, or both, indicating that the door is open.

[0078] In some examples, the control system **400** receives sensor data indicating that the door is propped open to a hold-open position. The hold-open position can be a door position in which the door remains open at a fixed swing angle. The door **100** being at the hold-open position can permit personnel to perform restocking, maintenance, or cleaning of the internal volume of the display case **11**, without the door **100** interfering with the personnel. In some examples, the sensor data indicating that the door **100** is in the hold-open position includes proximity sensor data indicating a distance between the proximity sensor **68b** and a door frame. In some examples, the sensor data indicating that the door **100** is in the hold-open position includes contact sensor data generated by a contact sensor **68a**. The contact sensor **68a** can be attached to a door closer of the door **100**. The

contact sensor data can detect when an interlock of the door closer is engaged in the hold-open position.

[0079] In some examples, the control system **400** receives sensor data from the sensors **68** continuously, near continuously, intermittently, or periodically. For example, the sensors **68** can output sensor data to the control system **400** at designated time intervals. The sensor data can indicate a current status of the door **100** (e.g., an open/shut status, a swing angle, an amount of time that the door has been open or shut).

[0080] In some examples, the control system **400** receives sensor data from the sensors **68** when the sensors **68** detect a change in a status of the door **100**. For example, the sensors **68** can output sensor data to the control system **400** when the sensors **68** detect a change from the door **100** being open to the door **100** being shut.

[0081] In some examples, the control system **400** receives sensor data from the sensors **68** when the sensors **68** detect that a time threshold has been exceeded. For example, the sensors **68** can output sensor data to the control system **400** when the sensors **68** detect that the door has been open for a threshold open time duration or has been shut for a threshold shut time duration.

[0082] The process **800** includes determining whether the door is open for a threshold open time duration (**804**). The threshold open time duration can be, for example, a specified number of seconds or minutes. The door **100** may be open for the threshold open time duration, for example, when a customer or employee forgets to shut the door **100**, or when the display case is being restocked or cleaned.

[0083] In some examples, the threshold open time duration is a time that corresponds to viewing time durations by customers, such as average viewing time durations by customers or maximum viewing time durations by customers. For example, an average viewing time by customers when viewing items inside the display case may be thirty seconds. The threshold open time duration can therefore be set to a time that matches or exceeds thirty seconds (e.g., forty-five seconds, sixty seconds, ninety-seconds). In some examples, the threshold open time duration is a time that corresponds to restocking time durations, such as minimum restocking time durations or average restocking time durations. For example, a minimum restocking time duration for personnel restocking the display case may be two minutes. The threshold open time duration can therefore be set to a time that matches or exceeds two minutes.

[0084] The process **800** includes, in response to determining that the door is open for the threshold time duration, adjusting display power to a reduced power setting (**806**). For example, the control system **400** can determine that the door **100** is open for at least the threshold time open duration. In response, the control system **400** can adjust power of the display **110** to a reduced power setting. In some examples, the control system **400** can adjust the power setting of the display **110** to a reduced power setting by transmitting an instruction to the circuitry module **526** to present no content on the display **110**. In some examples, adjusting display power to the reduced power setting includes turning the display **110** off. In some examples, adjusting display power to the reduced power setting includes setting the display **110** to a sleep mode. In some examples, adjusting display power to the reduced power setting includes reducing a display brightness of the display **110**.

[0085] The process **800** includes, in response to determining that the door is not open for the threshold open time duration, making no change to the display power (**808**). In some examples, the control system **400** can measure an amount of time that the door is open using a timer. When the door is shut before expiration of the threshold open time duration, the control system **400** can reset the timer. While the door is open and the threshold open time duration has not expired, the control system **400** makes no change to the display power. Therefore, while customers open and shut the door **100** to view and access items inside the display case, the display power remains on at full power. This reduces the amount of power cycling of the display **110**. This also enables a customer to view the display **110** while another customer has opened the door **100** to access the internal volume of the display case **11**.

[0086] The process **800** includes receiving data indicating that the door of the refrigerated display case is shut (**810**). For example, the control system **400** can receive data from the sensor **68**, the camera **65**, or both, indicating that the door is shut.

[0087] The process **800** includes adjusting display power to a non-reduced power setting (**812**). For example, in response to receiving the data indicating that the door **100** is shut, the control system **400** can adjust the display power to a non-reduced power setting. In some examples, adjusting the display power to the non-reduced power setting includes turning the display **110** on. In some examples, adjusting display power to the non-reduced power setting includes waking the display **110** from a sleep mode. In some examples, adjusting display power to the reduced power setting includes increasing a display brightness of the display **110**.

[0088] By adjusting a display power setting based on the open/shut status of the movable door, power consumption of the refrigerated display case **11** can be reduced while reducing impact to customer experience. For example, the display **110** can remain powered on at full power while customers open and shut the door to view and obtain items inside the display case **11**. When the door **100** is open for the threshold open time duration, such as during maintenance, restocking, or cleaning, the power setting of the display **110** can be reduced in order to conserve energy.

[0089] FIG. **9** shows a flow chart of an example process **900** for reducing power consumption of a refrigerated display case with an electronic door display by adjusting a lighting setting. The process **900** can be performed by a computing system including one or more computers, e.g., control system **400**. The order of steps in the process **900** is illustrative only, and the steps can be performed in different orders and/or in parallel. In some implementations, the process **900** can include additional steps, fewer steps, or some of the steps can be divided into multiple steps. In some examples, the steps of the process **900** can be performed by different components of the disclosed systems.

[0090] The process **900** includes receiving data indicating that a door of a refrigerated display case is open (**902**). For example, the control system **400** can receive data from the sensor **68**, the camera **65**, or both, indicating that the door is open.

[0091] The process **900** includes turning lighting on (**904**). For example, the control system **400** can adjust a power setting of lighting by turning on the lights **64** inside the display case **11**.

[0092] The process **900** includes receiving data indicating that the door of the refrigerated display case is shut (**906**). For example, the control system **400** can receive data from the sensor **68**, the camera **65**, or both, indicating that the door is shut.

[0093] The process **900** includes determining whether the door is shut for a threshold shut time duration (**908**). The threshold shut time duration can be, for example, a time delay of a specified number of seconds or minutes. The door **100** may be shut for the threshold shut time duration, for example, when a customer or employee shuts the door **100** and time passes before a next customer or employee opens the door **100**.

[0094] The process **900** includes, in response to determining that the door is shut for the threshold shut time duration, turning lighting off (**910**). For example, the control system **400** can determine that the door **100** is shut for at least the threshold time shut duration. In response, the control system **400** can turn off the lights **64**. The time delay can permit the camera **65** to obtain images of the internal volume of the display case while the lights are still on, prior to the lights turning off. Additionally, the time delay can prevent the lights from cycling too frequently, such as when the door **100** is opened and shut repeatedly over a short period of time.

[0095] The process **900** includes, in response to determining that the door is not shut for the threshold time duration, keeping lighting on (**912**). In some examples, the control system **400** can measure an amount of time that the door is shut using a timer. When the door is opened before expiration of the threshold shut time duration, the control system **400** can reset the timer. While the door is shut and the threshold shut time duration has not expired, the control system **400** makes no change to the lighting. Therefore, the lights **64** can remain powered on in between a first customer

shutting the door **100** and a next customer opening the door **100**. Keeping the lights **64** on can reduce cycling of the lights **64** on and off.

[0096] By adjusting lighting based on the open/shut status of the movable door, power consumption of the refrigerated display case **11** can be reduced while reducing impact to customer experience. For example, the lights **64** can be turned on when customers open the door to view and obtain items inside the display case **11**. When the door **100** is shut for the threshold shut time duration, such as in between customers accessing the display case **11**, the lights **64** can be turned off order to conserve energy.

[0097] In some examples, the control system **400** can adjust power settings of the lights **64** to reduce power consumption by adjusting a brightness setting of the lights **64**. For example, instead of turning the lights **64** off and on, the control system **400** can adjust the lights to a dimmed brightness and then to a non-dimmed brightness. In some examples, the control system **400** can control multiple lights **64** independently. For example, the control system **400** can control the light **64a** separately from the light **64b**. The control system **400** can therefore adjust lighting brightness within the display case to reduce power consumption by turning off some lights **64** while leaving other lights on.

[0098] FIG. **10** shows a flow chart of an example process **1000** for reducing power consumption of a refrigerated display case with an electronic door display by adjusting a heater setting. The process **1000** can be performed by a computing system including one or more computers, e.g., control system **400**. The order of steps in the process **1000** is illustrative only, and the steps can be performed in different orders and/or in parallel. In some implementations, the process **1000** can include additional steps, fewer steps, or some of the steps can be divided into multiple steps. In some examples, the steps of the process **1000** can be performed by different components of the disclosed systems.

[0099] The process **1000** includes receiving data indicating that a door of a refrigerated display case is open (**1002**). For example, the control system **400** can receive data from the sensor **68**, the camera **65**, or both, indicating that the door is open.

[0100] The process **1000** includes determining whether the door is open for a threshold open time duration (**1004**). The threshold open time duration can be, for example, a specified number of seconds or minutes. The door **100** may be open for the threshold open time duration, for example, when a customer or employee forgets to shut the door **100**, or when the display case **11** is being restocked or cleaned. In some examples, the threshold open time duration of the process **1000** is the same as the threshold open time duration of the process **800**. In some examples, the threshold open time duration of the process **1000** is different from the threshold open time duration of the process **800**.

[0101] The process **1000** includes, in response to determining that the door is open for the threshold open time duration, turning a heater off (**1006**). For example, the control system **400** can determine that the door **100** is open for at least the threshold time open duration. In response, the control system **400** can adjust a power setting of the heater **66** by turning off the heater **66**. In some examples, adjusting a power setting of the heater **66** includes reducing a heat output of the heater **66** to a reduced heat output.

[0102] The process **1000** includes, in response to determining that the door is not open for the threshold time open duration, keeping the heater on (**1008**). In some examples, the control system **400** can measure an amount of time that the door is open using a timer. When the door is shut before expiration of the threshold open time duration, the control system **400** can reset the timer. While the door is open and the threshold open time duration has not expired, the control system **400** makes no change to the heater **66**. Therefore, while customers open and shut the door **100** to view and access items inside the display case **11**, the heater remains on. This reduces the amount of power cycling of the heater **66**.

[0103] The process **1000** includes receiving data indicating that the door of the refrigerated display

case is shut (**1010**). For example, the control system **400** can receive data from the sensor **68**, the camera **65**, or both, indicating that the door is shut.

[0104] The process **1000** includes determining whether the door is shut for a threshold shut time duration (**1012**). The threshold shut time duration can be, for example, a specified number of seconds or minutes. The door **100** may be shut for the threshold open time duration, for example, when a customer or employee shuts the door **100** and time passes before a next customer or employee opens the door **100**. In some examples, the threshold shut time duration of the process **1000** is the same as the threshold shut time duration of the process **900**. In some examples, the threshold shut time duration of the process **1000** is different from the threshold shut time duration of the process **900**.

[0105] The process **1000** includes, in response to determining that the door is shut for the threshold shut time duration, turning the heater on (**1014**). For example, the control system **400** can determine that the door **100** is shut for at least the threshold time shut duration. In response, the control system **400** can turn on the heater **66**.

[0106] The process **1000** includes, in response to determining that the door is not shut for the threshold shut time duration, keeping the heater off (**1016**). In some examples, the control system **400** can measure an amount of time that the door **100** is shut using a timer. When the door is opened before expiration of the threshold shut time duration, the control system **400** can reset the timer. While the door is shut and the threshold shut time duration has not expired, the control system **400** makes no change to the heater **66**. Therefore, the heater **66** can remain powered off in between a first customer shutting the door **100** and a next customer opening the door **100** before the threshold shut time duration expires. Keeping the heater **66** off can reduce cycling of the heater **66** on and off.

[0107] By adjusting operation of the heater **66** based on the open/shut status of the movable door, power consumption of the refrigerated display case can be reduced while reducing impact to customer experience. For example, the heater **66** can remain powered on while customers open and shut the door to view and obtain items inside the display case, reducing condensation in the door **100**. When the door **100** is open for the threshold open time duration, such as during maintenance, restocking, or cleaning, the heater **66** can be turned off in order to conserve energy.

[0108] Implementations of the subject matter and the operations described in this specification can be realized in analog or digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. Implementations of the subject matter described in this specification can be realized using one or more computer programs, i.e., one or more modules of computer program instructions, encoded on computer storage medium for execution by, or to control the operation of, data processing apparatus. A computer storage medium can be, or be included in, a computer-readable storage device, a computer-readable storage substrate, a random or serial access memory array or device, or a combination of one or more of them. Moreover, while a computer storage medium is not a propagated signal; a computer storage medium can be a source or destination of computer program instructions encoded in an artificially generated propagated signal. The computer storage medium can also be, or be included in, one or more separate physical components or media (e.g., multiple CDs, disks, or other storage devices).

[0109] The operations described in this specification can be implemented as operations performed by a data processing apparatus on data stored on one or more computer-readable storage devices or received from other sources.

[0110] The term “data processing apparatus” encompasses all kinds of apparatus, devices, and machines for processing data, including by way of example a programmable processor, a computer, a system on a chip, or multiple ones, or combinations, of the foregoing. The apparatus can include special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). The apparatus can also include, in addition to hardware,

code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, a cross-platform runtime environment, a virtual machine, or a combination of one or more of them. The apparatus and execution environment can realize various different computing model infrastructures, such as web services, distributed computing and grid computing infrastructures. [0111] A computer program (also known as a program, software, software application, script, or code) can be written in any form of programming language, including compiled or interpreted languages, declarative or procedural languages, and it can be deployed in any form, including as a stand alone program or as a module, component, subroutine, object, or other unit suitable for use in a computing environment. A computer program can, but need not, correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

[0112] The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform actions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatus can also be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit).

[0113] Devices suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; [0114] magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

[0115] Thus, particular implementations of the subject matter have been described. Other implementations are within the scope of the following claims. In some cases, the actions recited in the claims can be performed in a different order and still achieve desirable results. In addition, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In certain implementations, multitasking and parallel processing can be advantageous.

[0116] While a number of examples have been described for illustration purposes, the foregoing description is not intended to limit the scope of the invention, which is defined by the scope of the appended claims. There are and will be other examples and modifications within the scope of the following claims. Furthermore, one of skill in the art would appreciate that features described in reference to a specific embodiment are not limited to that embodiment and can be interchanged with features of other embodiments.

Claims

1. A refrigerated display case, comprising: a movable door configured to provide access to the refrigerated display case; an electronic display mounted to the movable door; a sensor configured to indicate one or more positions of the movable door; at least one processor in electronic communication with the sensor and the electronic display; and a data store coupled to the at least one processor having instructions stored thereon which, when executed by the at least one processor, causes the at least one processor to perform operations comprising: receiving, from the sensor, data indicating a position of the movable door; and in response to receiving the data indicating the position of the movable door, adjusting a power setting of the electronic display.

2. The refrigerated display case of claim 1, wherein adjusting the power setting of the electronic display comprises turning the electronic display off or on.
3. The refrigerated display case of claim 1, comprising: a circuitry module comprising a media player in electronic communication with the electronic display, the media player configured to control media content presented on the electronic display.
4. The refrigerated display case of claim 3, wherein adjusting the power setting of the electronic display comprises transmitting an instruction to the circuitry module to present no content on the electronic display.
5. The refrigerated display case of claim 1, comprising a light configured to illuminate an interior of the refrigerated display case, the operations comprising: in response to receiving the data indicating the position of the movable door, adjusting a power setting of the light.
6. The refrigerated display case of claim 5, wherein adjusting the power setting of the light comprises turning the light off or on.
7. The refrigerated display case of claim 5, wherein: the data indicating the position of the movable door comprises data indicating that the movable door is shut; and adjusting the power setting of the light comprises turning the light off after a delay period.
8. The refrigerated display case of claim 1, comprising a heater configured to provide heat to at least a part of the refrigerated display case, the operations comprising: in response to receiving the data indicating the position of the movable door, adjusting a power setting of the heater.
9. The refrigerated display case of claim 8, wherein adjusting the power setting of the heater comprises turning the heater off or on.
10. The refrigerated display case of claim 8, wherein the heater comprises an anti-condensation heater configured to provide heat to the movable door.
11. The refrigerated display case of claim 1, wherein the data indicating the position of the movable door includes data indicating that the movable door is open or shut.
12. The refrigerated display case of claim 1, wherein the data indicating the position of the movable door includes data indicating that the movable door has been open for at least a threshold time duration.
13. The refrigerated display case of claim 1, wherein the data indicating the position of the movable door includes data indicating a swing angle of the movable door.
14. The refrigerated display case of claim 1, wherein an internal volume of the refrigerated display case holds an inventory, and the electronic display is configured to present a planogram representing the inventory.
15. The refrigerated display case of claim 1, wherein the movable door comprises an insulated panel.
16. The refrigerated display case of claim 1, wherein the sensor comprises at least one of a camera, a proximity sensor, a contact sensor, a position switch, a rheostat, an accelerometer, a gyroscope, an infrared sensor, a radar sensor, and an ultrasonic sensor.
17. A refrigerated display case system, comprising: a case including an internal volume; and a door system comprising: a movable door configured to provide access to the internal volume of the case; an electronic display mounted to the movable door; a sensor configured to indicate one or more positions of the movable door; at least one processor in electronic communication with the sensor and the electronic display; and a data store coupled to the at least one processor having instructions stored thereon which, when executed by the at least one processor, causes the at least one processor to perform operations comprising: receiving, from the sensor, data indicating a position of the movable door; and in response to receiving the data indicating the position of the movable door, adjusting a power setting of the electronic display.
18. The refrigerated display case system of claim 17, wherein adjusting the power setting of the electronic display comprises turning the electronic display off or on.
19. The refrigerated display case system of claim 17, comprising: a circuitry module comprising a

media player in electronic communication with the electronic display, the media player configured to control media content presented on the electronic display.

20. A computer-implemented method of operating a refrigerated display case, comprising: receiving, by one or more computers and from a sensor, data indicating a position of a movable door configured to provide access to the refrigerated display case; and in response to receiving the data indicating the position of the movable door, adjusting, by the one or more computers, a power setting of an electronic display mounted to the movable door.
